plot_spatial_anova_map_panel.R

August 10, 2026

Contents

`plot_spatial_anova_map_panel`
Build one spatial ANOVA map panel

Description

Builds a single `ggplot2` map panel showing the Shapley-adjusted percentage variance explained by Associations for one combination of continent, spatial scale, and taxonomic resolution. Grid rectangles are coloured by association percentage (viridis gradient, grey when no result). Fossil core locations are overlaid as points when available.

Usage

```
plot_spatial_anova_map_panel(
  sel_continent,
  sel_scale,
  sel_resolution,
  xlim_val,
  ylim_val,
  data_grid_with_results,
  data_coords_per_unit,
  sf_world,
  graphical_options,
  show_x_axis = TRUE,
  show_y_axis = TRUE,
  col_label = NULL,
  row_label = NULL
)
```

Arguments

- `sel_continent` Character scalar. Continent identifier (e.g. "europe", "america", "asia").
- `sel_scale` Character scalar. Spatial scale identifier (e.g. "continental", "regional", "local").
- `sel_resolution` Character scalar. Taxonomic resolution identifier (e.g. "genus", "family", "functional_type").
- `xlim_val` Numeric vector of length 2. Longitude limits for ggplot2::coord_sf() (western, eastern).
- `ylim_val` Numeric vector of length 2. Latitude limits for ggplot2::coord_sf() (southern, northern).
- `data_grid_with_results` Data frame. Spatial grid joined with ANOVA association percentages; must contain columns continent, scale, taxonomic_resolution, x_min, x_max, y_min, y_max, assoc_pct.
- `data_coords_per_unit` Data frame. Per-unit fossil core coordinates; must contain columns continent, scale, coords (list-column of data frames with coord_long and coord_lat).
- `sf_world` sf object. World polygon basemap used as the background layer.
- `graphical_options` Named list of canvas settings (e.g. from load_active_config_value("graphical")); must contain width, height, units, dpi, bg.
- `show_x_axis` Logical scalar. When FALSE, x-axis text and tick marks are hidden. Default TRUE.
- `show_y_axis` Logical scalar. When FALSE, y-axis text and tick marks are hidden. Default TRUE.
- `col_label` Character scalar or NULL. When not NULL, added as a bold column-header title above the panel via ggplot2::ggtitle(). Default NULL.
- `row_label` Character scalar or NULL. When not NULL, added as a bold rotated y-axis title for the panel row. Default NULL.

Details

The function applies ggview::canvas() using the supplied graphical_options and ends with ggplot2::coord_sf() so that the continent-specific bounding box takes precedence over the automatic bounding box added by ggplot2::geom_sf().

Value

A ggplot object representing one map panel.

See Also

extract_jsdm_variance_fractions, compute_shapley_variance_components

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